A kind of method for preparing 7 hydroxyl mitragynines

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Abstract

translated from Chinese

The invention discloses a convenient method for synthesizing 7-hydroxy mitragynine, which belongs to the technical field of medicine. It uses mitragynine as the starting material, tetrahydrofuran and water as the solvent, and reacts with [bis(trifluoroacetoxy)iodo]benzene in an argon atmosphere at 0°C, and passes the obtained crude product through Purify by basic column chromatography to prepare and obtain the 7-hydroxy mitragynine. The materials required by the invention are easy to obtain, low in cost, simple in reaction operation and easy to handle, and can obtain a large amount of 7-hydroxy mitragynine for use in medical research and development.

Classifications

C07D471/14 Ortho-condensed systems

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Claims (5)

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1. a method for preparing 7-hydroxy mitragynine, is characterized in that: with tetrahydrofuran and water as solvent, taking mitragynine as starting raw material, with [bis(trifluoroacetoxy) iodine] After benzene reacts at low temperature, it is purified by basic column chromatography to prepare the 7-hydroxy mitragynine. 2. the method for preparing 7-hydroxy mitragynine according to claim 1, is characterized in that: comprise the following steps: 1) Dissolve mitragynine in a mixture of tetrahydrofuran and water, then add [bis(trifluoroacetoxy)iodo]benzene in an argon atmosphere at 0~2 °C, mix well and react at low temperature for 3 ~8 hours; 2) Add saturated sodium bicarbonate solution after the reaction, adjust the pH of the reaction solution to 8~9, then add ethyl acetate to wash, then extract with saturated saline three times, combine the extracts, add anhydrous sodium sulfate to dry, After spin-drying, dark brown crude product was obtained; 3) Purify the obtained crude product by basic column chromatography to obtain high-purity 7-hydroxy-mitragynine. 3. The method for preparing 7-hydroxymitragynine according to claim 2, characterized in that: Step 1) The volume ratio of tetrahydrofuran to water in the mixed solution is 2:1-2. 4. The method for preparing 7-hydroxy mitragynine according to claim 2, characterized in that: the weight ratio of mitragynine used in step 1) to [bis(trifluoroacetoxy) iodine]benzene is 5 :8~5:11. 5. the method for preparing 7-hydroxy mitragynine according to claim 2, is characterized in that: the basic column chromatography described in step 3) is to adopt basic silica gel column, with the ethyl acetate of volume ratio 1:1 The ester-petroleum ether mixture is used as the eluent for separation and purification; The basic silica gel column is obtained by washing an ordinary silica gel column with triethylamine and petroleum ether in sequence.

Description

translated from Chinese

A kind of method for preparing 7-hydroxy mitragynine

technical field

The invention belongs to the technical field of medicine, and in particular relates to a method for preparing 7-hydroxy mitragynine.

Background technique

Kratom (kratom) is native to India and Malaysia in Southeast Asia. Kratom leaves are often used as a substitute for opium and can be eaten by smoking, chewing or making tea. Low-dose kratom has a stimulating effect. For a long time, many manual laborers in Southeast Asia have used kratom as a stimulant to offset work fatigue. High-dose kratom has a sedative and anesthetic effect and is used as a drug-detoxification substitute for opium or for treatment Opiate withdrawal symptoms. Kratom has recently become more widely recognized and used due to its valuable application in many conditions and diseases such as fever, pain, diarrhea, cough, hypertension, depression, anxiety and withdrawal from opioid addiction. More than 40 kinds of alkaloids have been isolated from kratom leaves, among which Mitragynine and 7-Hydroxymitragynine have been proved to have good biological activity.

Mitraphylline is a central nervous system stimulant, which has analgesic effects similar to morphine, but its mechanism of action is completely different from that of morphine. Currently, preclinical studies have used mitragynine as an analgesic, anti-inflammatory, potent opioid receptor agonist. In Thailand, mitragynine is often used as an alternative to opioids for pain management. Studies have found that the opioid receptor activity of 7-hydroxy mitragynine is higher than that of mitragynine, and its analgesic effect is far stronger than that of morphine. 7-Hydroxymitragynine has high lipophilicity and is easy to pass through the blood-brain barrier. Its analgesic effect is faster and more potent than that of mitragynine and morphine. It is well known that oral morphine will be rapidly metabolized by the liver, and frequent injections of opioids such as morphine and codeine can cause serious side effects, such as respiratory depression, constipation, sedation, nausea, itching, etc. However, studies have found that oral administration of 7-hydroxycap Pylphylline can be well absorbed by the human body, and its metabolism in the body is more effective than morphine, and its side effects are far less

than morphine. A large number of research results show that the mechanism of action of 7-hydroxymitragynine is different from that of opioids, and its side effects are far less than that of morphine. 7-hydroxymitragynine is likely to become a new type of drug because of its unique structure and strong physiological activity. Analgesics, so 7-hydroxy mitragynine has great development significance. However, the content of 7-hydroxymitragynine in kratom leaves is very low, and the 7-hydroxymitragynine extracted from kratom leaves only accounts for about 2%, so the production cost of extracting from plants is high, which seriously affects the Pharmacodynamic evaluation of the compound in vitro and in vivo.

Researchers at home and abroad have adopted a variety of methods to obtain 7-hydroxymitragynine. Although the existing methods can synthesize the target product, these methods still have some disadvantages, such as complex synthesis steps, many by-products, low conversion rate, Especially in the purification process, factors such as the extremely low yield due to the failure to find suitable purification conditions are not conducive to large-scale preparation. Therefore, whether a large amount of 7-hydroxymitragynine can be synthesized by finding suitable chemical synthesis conditions and designing a simple chemical synthesis route has become an urgent problem to be solved.

Mitragynine is an alkaloid extracted and isolated from kratom leaves. It is the main active ingredient of kratom leaves, and its content accounts for more than 60% of the total alkaloid extract. Its source is very rich and the price is cheap. Our research group found that in the process of synthesizing 7-hydroxymitragynine, if [bis(trifluoroacetoxy)iodo]benzene (PIFA) was added to mitragynine under argon atmosphere with tetrahydrofuran and water as solvents, In alkali and placed in a low temperature environment for reaction, the 7-hydroxy mitragynine with a yield of about 70% can be obtained with a conversion rate of more than 95% and a common silica gel chromatography column treated with triethylamine. Therefore, using mitragynine

as a starting material to synthesize 7-hydroxymitragynine can effectively solve the problem of insufficient raw materials for 7-hydroxymitragynine, and can also provide a new solution for the synthesis of other similar compounds ideas and plans.

Contents of the invention

The present invention aims at the problem that there are many difficulties in the preparation and extraction of 7-hydroxy mitragynine by various existing methods, and provides a simple method for preparing 7-hydroxy mitragynine, which can greatly improve the extraction efficiency of 7-hydroxy mitragynine. The yield of streptine can reduce the production cost.

To achieve the above object, the present invention adopts the following technical solutions:

A method for preparing 7-hydroxy mitragynine, which uses tetrahydrofuran and water as solvents, and uses mitragynine as a starting material, and reacts it with [bis(trifluoroacetoxy)iodo]benzene at low temperature Afterwards, through basic column chromatography purification, preparation obtains described 7-hydroxy mitragynine; It comprises the following steps:

1) Dissolve mitragynine in a mixture of tetrahydrofuran and water, then add [bis(trifluoroacetoxy)iodo]benzene (PIFA) in an argon atmosphere at 0-2 °C, and mix well Low temperature reaction for 3~8 hours;

2) Add saturated sodium bicarbonate solution after the reaction, adjust the pH of the reaction solution to 8~9, then add ethyl acetate to wash, then extract with saturated saline three times, combine the extracts, add anhydrous sodium sulfate to dry, After spin-drying, dark brown crude product was obtained;

3) Purify the obtained crude product by basic column chromatography to obtain high-purity 7-hydroxy-mitragynine.

Step 1) The volume ratio of tetrahydrofuran to water in the mixed solution is 2:1-2; the weight ratio of

mitragynine to [bis(trifluoroacetoxy)iodo]benzene used is 5:8~5:11.

The basic column chromatography described in step 3) uses a basic silica gel column, and the ethyl acetate-petroleum ether mixture with a volume ratio of 1:1 is used as an eluent for separation and purification; the basic silica gel column is made of ordinary The silica gel column was washed successively with triethylamine and petroleum ether.

The remarkable advantage of the present invention is that: aiming at the problems of low yield and high extraction cost in the preparation and extraction of 7-hydroxy mitragynine by existing methods, the present invention provides a simple and convenient preparation method of 7-hydroxy mitragynine, It directly uses mitragynine as the starting material for synthesis, and the reagents used are commonly used reagents in the laboratory. Therefore, the materials required by the present invention are easy to obtain, low in cost, simple in reaction operation, easy to handle, and high in purification yield. 7-Hydroxymitragynine can be obtained in large quantities for use in pharmaceutical research and development.

Description of drawings

Figure 1 is a schematic diagram of the synthesis of the present invention.

detailed description

In order to make the content of the present invention easier to understand, the technical solutions of the present invention will be further described below in conjunction with specific embodiments, but the present invention is not limited thereto.

Rinse an ordinary silica gel column containing 50 grams of silica gel with 20-30 mL of triethylamine and then with 50 mL of petroleum ether to obtain an alkaline silica gel column.

Example 1

1) Add 20 mg of mitragynine, 1 mL of tetrahydrofuran and 0.5 mL of water into a 100 mL flask, fill it with argon, then place the flask in a low-temperature reactor and stir; another 32 mg of [bis(trifluoroacetoxy) Iodo]benzene (PIFA) was dissolved in 1 mL of tetrahydrofuran, then added dropwise into the flask at 0 °C, and reacted for 4 hours;

2) Step 1) After the reaction, add 2 mL of saturated sodium bicarbonate solution to adjust the pH of the reaction solution to 8~9, then add 20 mL of ethyl acetate to wash, then extract with saturated saline three times, 15 mL each time, and combine the extractions liquid, then add anhydrous sodium sulfate to dry and filter, and the obtained filtrate is spin-dried to obtain a dark brown crude product;

3) Using the ethyl acetate-petroleum ether mixture with a volume ratio of 1:1 as the eluent, the obtained crude product was separated and purified by a basic silica gel column to obtain a light yellow solid with a yield of 50%.

Example 2

1) Add 20 mg mitragynine, 1 mL tetrahydrofuran and 0.5 mL water into a 100 mL flask, fill with argon, then place the flask in a low-temperature reactor and stir; another 32 mg [bis(trifluoroacetoxy Base) iodo]benzene (PIFA) was dissolved in 1 mL of tetrahydrofuran, then added dropwise into the flask at 0 °C, and reacted for 6 hours;

2) Step 1) After the reaction, add 2 mL of saturated sodium bicarbonate solution to adjust the pH of the reaction solution to 8~9, then add 20 mL of ethyl acetate to wash, then extract with saturated saline three times, 15 mL each time, and combine the extractions liquid, then add anhydrous sodium sulfate to dry and filter, and the obtained filtrate is spin-dried to obtain a dark brown crude product;

3) Using the ethyl acetate-petroleum ether mixture with a volume ratio of 1:1 as the eluent, the obtained crude product was separated and purified by a basic silica gel column to obtain a light yellow solid with a yield of 65%.

Example 3

1) Add 20 mg of mitragynine, 1 mL of tetrahydrofuran and 0.5 mL of water into a 100 mL flask, fill with argon, then place the flask in a low-temperature reactor and stir; another 32 mg of [bis(trifluoroacetoxy) Iodo]benzene (PIFA) was dissolved in 1 mL of tetrahydrofuran, then added dropwise into the flask at 0 °C, and reacted for 8 hours;

2) Step 1) After the reaction, add 2 mL of saturated sodium bicarbonate solution to adjust the pH of the reaction solution to 8~9, then add 20 mL of ethyl acetate to wash, then extract with saturated saline three times, 15 mL each time, and combine the extractions liquid, then add anhydrous sodium sulfate to dry and filter, and the obtained filtrate is spin-dried to obtain a dark brown crude product;

3) Using ethyl acetate-petroleum ether mixture with a volume ratio of 1:1 as the eluent, the obtained crude product was separated and purified by basic silica gel column to obtain a light yellow solid with a yield of 70%.

Example 4

1) Add 20 mg of mitragynine, 1 mL of tetrahydrofuran and 0.5 mL of water into a 100 mL flask, fill it with argon, then place the flask in a low-temperature reactor and stir; another 43 mg of [bis(trifluoroacetoxy Base) iodo]benzene (PIFA) was dissolved in 1 mL of tetrahydrofuran, then added dropwise into the flask at 2 °C, and reacted for 3 hours;

2) Step 1) After the reaction, add 2 mL of saturated sodium bicarbonate solution to adjust the pH of the reaction solution to 8~9, then add 20 mL of ethyl acetate to wash, then extract with saturated saline three times, 15 mL each time, and combine the extractions liquid, then add anhydrous sodium sulfate to dry and filter, and the obtained filtrate is spin-dried to obtain a dark brown crude product;

3) Using the ethyl acetate-petroleum ether mixture with

a volume ratio of 1:1 as the eluent, the obtained crude product was separated and purified by a basic silica gel column to obtain a light yellow solid with a yield of 75%.

Example 5

1) Add 100 mg of mitragynine, 2 mL of tetrahydrofuran and 1 mL of water into a 100 mL flask, fill it with argon, then place the flask in a low-temperature reactor and stir; another 215 mg of [bis(trifluoroacetoxy Base) iodo]benzene (PIFA) was dissolved in 2 mL of tetrahydrofuran, then added dropwise into the flask at 2 °C, and reacted for 3 hours;

2) Step 1) After the reaction, add 4 mL of saturated sodium bicarbonate solution to adjust the pH of the reaction solution to 8~9, then add 30 mL of ethyl acetate to wash, then extract with saturated saline three times, 15 mL each time, and combine the extractions liquid, then add anhydrous sodium sulfate to dry and filter, and the obtained filtrate is spin-dried to obtain a dark brown crude product;

3) Using ethyl acetate-petroleum ether mixture with a volume ratio of 1:1 as the eluent, the obtained crude product was separated and purified by basic silica gel column to obtain a light yellow solid with a yield of 70%.

The light yellow solids obtained in Examples 1-5 were determined by hydrogen spectrum ^{1} H-NMR, carbon spectrum ^{13} C-NMR, high-resolution mass spectrometry and high-performance liquid chromatography HPLC respectively, and it was determined that they were all 7-hydroxy mitragynine , and its content is greater than 95%, the relevant measurement data are as follows:

^{1} H-NMR (400 MHz, $CDCl_3$) δ: 7.45 (s, 1H), 7.28 (t, *J* = 7.9 Hz, 1H), 7.20 (d, *J* = 7.6Hz, 1H), 6.71 (d, *J* = 8.2 Hz, 1H), 3.85 (s, 3H), 3.82 (s, 3H), 3.68 (s, 3H), 3.16 (d, *J* =9.2Hz, 1H), 3.07 (d, *J* =11.5 Hz, 1H) , 3.03-2.98 (m, 1H), 2.87-2.77 (m, 2H), 2.66 (t, *J* =9.5Hz, 2H), 2.55-2.48 (m, 1H), 1.88 (d, *J* =13.6 Hz, 1H), 1.75-1.59 (m, 3H), 1.30-1.20 (m, 1H), 0.85 (t, *J* =7.3 Hz, 3H).

^{13}C-NMR (101 MHz, $CDCl_3$) δ: 184.39, 169.17, 160.63, 155.82, 154.87, 130.48, 126.51, 114.01, 111.11, 108.76, 80.81, 77.32, 77.00, 76.658, 61.676, 58.3 , 49.95, 40.39, 39.23, 35.56, 25.95, 18.83, 12.72.

HRMS (ESI) calcd for $C_{23}H_{30}N_2O_5$ $[M+H]^+$ m/z 415.2227, found 415.2227.

The above descriptions are only preferred embodiments of the present invention, and all equivalent changes and modifications made according to the scope of the patent application of the present invention shall fall within the scope of the present invention.

Patent Citations (1)

Publication number	Priority date	Publication date
WO2016176657A1 *	2015-04-30	2016-11-03
Family To Family Citations		

* Cited by examiner, † Cited by third party

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Priority And Related Applications

Priority Applications (1)

Application	Priority date	Filing date	Title
CN201710380436.1A	2017-05-25	2017-05-25	A kind of method fo preparing hydroxyl mitragyni

Applications Claiming Priority (1)

Application	Filing date	Title
CN201710380436.1A	2017-05-25	A kind of method for preparing 7 hydroxyl mitragynines

Legal Events

Date	Code	Title
2017-07-21	PB01	Publication
2017-08-15	SE01	Entry into force of request for substantive
2017-08-15	SE01	Entry into force of request for substantive
2020-01-10	RJ01	Rejection of invention patent application a
2020-01-10	RJ01	Rejection of invention patent application a